Problem Set 5 — Smoothing

Statistical Modelling (BAM), IE University — Spring 2027

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Instructions. Sessions 20–22 and Chapters 14–15 of the class notes. Work in pairs; each student submits their own PDF (rendered from this `.Rmd`) on Blackboard. Show derivations, give numbers to 3–4 significant digits, and interpret every number you report in context. Use `set.seed(2027)` whenever you simulate or shuffle. Total: 100 points (estimated effort 6–8 hours).

Notation as in the course: $\hat{\boldsymbol{f}} = S\boldsymbol{y}$, edf $= \mathrm{tr}(S)$, kernel $K_h(u) = K(u/h)/h$, P-spline $\hat{\theta} = (B^\top B + \lambda D_d^\top D_d)^{-1} B^\top y$. The following helper functions are given (read them: you will use and sometimes modify them).

```
library(faraway); library(splines); library(mgcv); library(MASS)
nw_S <- function(x0, x, h) {                          # Nadaraya-Watson, Gaussian kernel
  K <- dnorm(outer(x0, x, "-") / h); K / rowSums(K)
}
ll_S <- function(x0, x, h) {                          # local linear, Gaussian kernel
  t(vapply(seq_along(x0), function(i) {
    d <- x - x0[i]; w <- dnorm(d / h)
    s0 <- sum(w); s1 <- sum(w * d); s2 <- sum(w * d^2)
    w * (s2 - d * s1) / (s0 * s2 - s1^2)
  }, numeric(length(x))))
}
bbase <- function(x, xl = min(x) - 0.01, xr = max(x) + 0.01, nseg = 20, bdeg = 3) {
  dx <- (xr - xl) / nseg
  knots <- seq(xl - bdeg * dx, xr + bdeg * dx, by = dx)
  B <- splineDesign(knots, x, ord = bdeg + 1, outer.ok = TRUE); attr(B, "knots") <-
↪  knots; B
}
psfit <- function(B, y, lambda, d = 2) {
  D <- diff(diag(ncol(B)), differences = d)
  A <- crossprod(B) + lambda * crossprod(D); n <- length(y)
  theta <- solve(A, crossprod(B, y)); f <- drop(B %*% theta)
  edf <- sum(diag(solve(A, crossprod(B)))); rss <- sum((y - f)^2)
  list(theta = drop(theta), f = f, edf = edf, rss = rss, gcv = n * rss / (n - edf)^2,
       sigma = sqrt(rss / (n - edf)), A = A)
}
```

1 Local constant versus local linear (10 points)

Let $x = (0, 1, 2, 3)$, $y = (1, 2, 2, 5)$ and use the triangular kernel $K(u) = (1 - |u|)_+$ with $h = 2$ (weights 1 at distance 0, $\frac{1}{2}$ at distance 1, 0 at distance ≥ 2).

(a) Compute by hand the Nadaraya–Watson and the local linear estimates of f at $x_0 = 0$ and at $x_0 = 1.5$.

(b) The local linear estimate is $\hat{f}(x_0) = \sum_j l_j(x_0) y_j$. Derive l_j from the weighted normal equations and prove that $\sum_j l_j = 1$ and $\sum_j l_j (x_j - x_0) = 0$.

(c) Suppose the data lie exactly on the line $y = 2 + 3x$. Use (b) to show that the local linear estimate is exact at *any* x_0 and for *any* design, and compute the bias of the NW estimate at $x_0 = 0$ for this line.

(d) Using your answers, explain in three sentences why Nadaraya–Watson is biased near the boundary of the design, how this relates to the term $f'p'/p$ in its bias, and why the bias at $x_0 = 1.5$ in (a) is the

same for both estimators.

2 Linear smoothers and leave-one-out (12 points)

(a) *(Proof.)* Let $\hat{f}_h(x) = \sum_j K_h(x-x_j)y_j / \sum_j K_h(x-x_j)$ with fixed h and $S_{ij} = K_h(x_i-x_j)/\sum_l K_h(x_i-x_l)$. Prove directly that $y_i - \hat{f}^{(-i)}(x_i) = (y_i - \hat{f}_i)/(1-S_{ii})$, where $\hat{f}^{(-i)}$ is computed without observation i.

(b) Let $x = (0,1,2,3)$, $y = (1,4,2,5)$ and a uniform kernel with weights 1 for $|x_i - x_j| \le 1.5$ and 0 otherwise. Write S, compute $\hat{\boldsymbol{f}}$, edf, the four leave-one-out residuals by the shortcut, LOOCV and GCV. Verify the residuals of observations 1 and 2 by recomputing the fit without them.

(c) Compute $\mathrm{tr}(SS^\top)$ for this S and compare it with $\mathrm{tr}(S)$. Explain why they coincide here (hint: look at the entries in each row) and what this implies for the estimator $\hat{\sigma}^2 = \mathrm{RSS}/(n - \mathrm{edf})$ (use the formula for $\mathrm{E}\,\mathrm{RSS}$ of a linear smoother).

(d) Why do LOOCV and GCV differ in (b)? When would they be equal?

3 Polynomial regression on the Ann Arbor temperatures (12 points)

The data frame `faraway::aatemp` gives the annual mean temperature (`temp`, degrees Fahrenheit) in Ann Arbor, Michigan, for $n = 115$ years between 1854 and 2000 (some years are missing before 1881). We want to describe the long-term trend.

```
data(aatemp); d <- aatemp; n <- nrow(d)
c(n = n, range(d$year), missing_years = diff(range(d$year)) + 1 - n)
```

```
#>              n                              missing_years
#>            115           1854           2000             32
```

(a) Fit polynomials of degree $q = 1, \dots, 8$ with `poly(year, q)`. Tabulate RSS, AIC, BIC and the largest leverage. Which degree do AIC and BIC choose? Choose also by 5-fold cross-validation with shuffled folds (`set.seed(2027)`).

(b) Fit the raw polynomial `lm(temp ~ poly(year, 5, raw = TRUE))`. What happens, and why? Compute the condition number of the raw design for $q = 5$, using `year` as it is and after centring and scaling `year` to $[-1, 1]$. Verify that the fitted values of the orthogonal and of the rescaled-raw fit coincide.

(c) Predict the temperature in 2000, 2020 and 2050 with the degree-2 and the AIC-selected polynomial. Comment on the plausibility of the predictions and on what the leverage table in (a) says about the end of the series.

(d) In the output of `summary(lm(temp ~ poly(year, 5)))` the *estimates* of the coefficients of the lower-order terms are exactly the same as in the fit of degree 3, and their standard errors differ only slightly. Explain why using the properties of orthogonal polynomials. What do the t-statistics of the quadratic, cubic and quintic terms tell you, and is a sequence of such tests a good way to choose the degree?

4 Exact bias and variance with a known truth (14 points)

The data frame `faraway::exa` has $n = 256$ observations $y_i = m(x_i) + \varepsilon_i$, $x_i \in (0,1)$, and also contains the true mean function `m` evaluated at the x_i. We treat the error variance as known and equal to $\sigma^2 = \overline{(y-m)^2}$.

```
data(exa); x <- exa$x; y <- exa$y; m <- exa$m; n <- length(y)
sigma2 <- mean((y - m)^2); round(c(sigma2 = sigma2, sigma = sqrt(sigma2)), 4)
```

```
#> sigma2  sigma
#> 0.0977 0.3126
```

(a) For a linear smoother $\hat{\boldsymbol{f}} = S\boldsymbol{y}$ and the *fixed* truth $\boldsymbol{m}$, derive the pointwise decomposition $\mathrm{E}(\hat{f}_i - m_i)^2 = b_i^2 + v_i$ with $b_i = \{(S-I)\boldsymbol{m}\}_i$ and $v_i = \sigma^2 (SS^\top)_{ii}$.

(b) For the Gaussian-kernel Nadaraya–Watson smoother on the grid `h <- seq(0.005, 0.08, by = 0.0025)`, compute the average squared bias, average variance and average MSE as functions of h and plot them. Report h_{MSE}, the bandwidth that minimises the MSE, and the corresponding edf.

(c) Now pretend that you do not know m: choose h by LOOCV and by GCV using only $\boldsymbol{y}$. Report both bandwidths and compute the *true* MSE at the selected bandwidths. By what percentage does the true MSE exceed its minimum?

(d) Split the design points into *boundary* ($x < 0.1$ or $x > 0.9$) and *interior* points and find the MSE-optimal bandwidth separately in each region. What do you observe and how does it relate to the theory of the NW bias at the boundary?

5 Penalized least squares by hand, and its mixed-model form (12 points)

Let $B = I_3$ (three coefficients, one per observation), $y = (1, 4, 2)^\top$, first-order differences $D_1 = \begin{bmatrix} -1 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix}$ and $\lambda = 2$. The criterion is $\|y - \theta\|^2 + \lambda \|D_1 \theta\|^2$.

(a) Write $A = I + \lambda D_1^\top D_1$ and solve $A\hat{\theta} = y$ by hand.

(b) Compute $\text{edf} = \text{tr}(A^{-1})$ and compare with its limits as $\lambda \to 0$ and $\lambda \to \infty$. What does $\hat{\theta}$ tend to as $\lambda \to \infty$ and why?

(c) Which moments $\sum_j j^k \hat{\theta}_j$ are conserved for $d = 1$? Check $k = 0$ and show that $k = 1$ fails here.

(d) *(Proof.)* Let $M = B^\top B$ be positive definite and $P = D_d^\top D_d$. Show that $\text{edf}(\lambda) = \sum_k (1 + \lambda \gamma_k)^{-1}$ where $\gamma_k \geq 0$ are the eigenvalues of $M^{-1/2} P M^{-1/2}$, and deduce that the edf is strictly decreasing in λ.

(e) For $d = 1$ give G and $Z = B D_1^\top (D_1 D_1^\top)^{-1}$ of the mixed-model representation, with $\theta = G\beta + D_1^\top (D_1 D_1^\top)^{-1} b$. If a mixed-model fit gave $\hat{\sigma}^2 = 1.5$ and $\hat{\sigma}_b^2 = 0.75$, which λ would this correspond to? Verify numerically in R that Henderson's equations with that λ reproduce $\hat{\theta}$.

6 Choosing the smoothness: LOESS, smoothing splines and P-splines (14 points)

We now estimate the trend of the Ann Arbor temperatures (`aatemp`, as in Problem 3) nonparametrically.

(a) Fit quadratic LOESS with spans $0.2, 0.3, 0.5, 0.75$ and report `trace.hat` for each. Choose the span by 10-fold CV with *shuffled* folds (`set.seed(2027)`; use `loess.control(surface = "direct")` for prediction).

(b) Fit `smooth.spline(year, temp)` with its default (GCV) and with `cv = TRUE`, and also `df = 6`. Report the degrees of freedom and λ.

(c) Fit `gam(temp ~ s(year, bs = "ps", k = 20))` with `method = "GCV.Cp"` and `method = "REML"`. Report the edf.

(d) Plot the data and all fits (LOESS at the CV span, the three smoothing splines, the two P-splines). Plot also the GCV criterion of a hand-coded P-spline (`bbase(year, nseg = 30)`, i.e. 33 B-splines, $d = 2$) against $\log_{10} \lambda$ for $\log_{10} \lambda \in [-3, 6]$, and report the number of local minima, their edf, and the relative range $\{\max - \min\}/\min$ of GCV over the λ values with $6 \leq \text{edf} \leq 26$.

(e) The default `smooth.spline` fit has `df` ≈ 23 while the REML fit has ≈ 5. Using the plot of the GCV curve explain what happened, say which fit you would report as the long-term trend and why, and what the data (115 values of annual means with year-to-year noise) can and cannot determine.

7 A P-spline by hand, and inference on a change (14 points)

Use the Ann Arbor temperatures, a cubic B-spline basis `Bt <- bbase(year, nseg = 17)` (20 B-splines) and a second-order penalty, unless stated otherwise.

(a) Using `psfit()`, find by `uniroot()` the smoothing parameter $\hat{\lambda}$ with edf $= 6$. Report $\hat{\lambda}$, the RSS and $\hat{\sigma}^2 = \text{RSS}/(n - \text{edf})$.

(b) Verify numerically, for $d = 2$, the conservation of moments $\sum_i x_i^k \hat{f}_i = \sum_i x_i^k y_i$ for $k = 0, 1$ (use $x_i =$ year -1900 for numerical stability) and show that it fails for $k = 2$. Explain with the theorem on conservation of moments, including its conditions.

(c) We want to quantify two changes in the average temperature: $\Delta_1 = \hat{f}(1950) - \hat{f}(1900)$ and $\Delta_2 = \hat{f}(2000) - \hat{f}(1950)$. Write $\Delta = c^\top \hat{\theta}$ where c is the difference of the rows of the B-spline matrix evaluated at the two years (use `splineDesign(attr(Bt, "knots"), c(1900, 1950, 2000), ord = 4, outer.ok = TRUE)`). Derive $\mathrm{Var}(c^\top \hat{\theta})$ both in the Bayesian way, $\sigma^2 c^\top A^{-1} c$ with $A = B^\top B + \lambda D_2^\top D_2$, and in the frequentist way, $\sigma^2 c^\top A^{-1} B^\top B A^{-1} c$. Compute $\hat{\Delta}_1, \hat{\Delta}_2$ and approximate 95% intervals (± 2 se) with $\hat{\sigma}$ from (a). What do you conclude?

(d) Repeat (a) and (c) with $d = 1$ and $d = 3$ (same edf $= 6$). How much do $\hat{\Delta}_1, \hat{\Delta}_2$ depend on the order of the penalty? Which null space does each d correspond to, and what does it mean for the shape of the fit when the penalty is very large?

(e) Compare $\hat{\Delta}_1, \hat{\Delta}_2$ and their standard errors with those from `mgcv`: fit `gam(temp ~ s(year, bs = "ps", k = 20), method = "REML")` and use `predict(g, newdata, type = "lpmatrix")` with `g$Vp`. State briefly why the numbers are close but not identical.

8 Critique and communicate (12 points)

A draft report contains the following paragraph.

> *"We modelled the annual mean temperature in Ann Arbor with a polynomial of degree 8, because among all the degrees we tried it had the smallest residual sum of squares. As a check, we plotted the fit ± 2 residual standard deviations: this is a 95% confidence band for the true trend, and since 95% of the observations fall inside it, the model is validated. We also tried a P-spline with 6 degrees of freedom, but kept the polynomial because its RSS is smaller and its coefficients are easier to report. Finally, we used the polynomial to forecast the temperature in 2030."*

(a) List at least five errors or misconceptions in the paragraph and correct each in one or two sentences.

(b) Support your corrections with numbers computed in R: RSS, edf (number of coefficients for the polynomial, `trace.hat` for LOESS) and GCV of the degree-8 polynomial, of LOESS with span 0.3 and of the P-spline with edf $= 6$ from Problem 7; the fraction of observations within ± 2 residual standard deviations for the P-spline; the pointwise standard error range of the P-spline fit ($\hat{\sigma}\sqrt{(S_\lambda S_\lambda^\top)_{ii}}$, Bayesian version $\hat{\sigma}\sqrt{(S_\lambda)_{ii}}$ is also acceptable); and the polynomial's predictions for 2000 and 2030.

(c) Write a corrected summary of about 150 words for readers who are not statisticians, with one sentence on uncertainty and one on what the data cannot tell us.